

# Lecture 3

Last class

- ▷ Gradient Descent
- ▷ PL Inequality

Today

- ▷ Least squares
- ▷ Implicit bias

HW1 and Project Overview out today!

Overparameterized Linear Least Squares

Today we go back to a classical problem. Suppose we observe feature vectors and observed responses

$$x_1, \dots, x_n \in \mathbb{R}^d \quad \text{and} \quad y_1, \dots, y_n \in \mathbb{R}.$$

We aim to fit these input-output pairs via a linear function, i.e., we want to solve

$$(v) \quad f_n(w) = \frac{1}{2n} \sum_{i=1}^n (w^T x_i - y_i)^2 = \frac{1}{2n} \|Xw - y\|_2^2.$$

$X = \begin{pmatrix} -x_1- \\ \vdots \\ -x_n- \end{pmatrix}$        $y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

We assume that

$$d > n$$

More parameters than observations

and

$$\text{rank}(X) = n.$$

Independent inputs.

Define

$$S^* = \{w \in \mathbb{R}^d \mid Xw = y\} \quad \text{and}$$

$$f^* = \min f_n(w) = 0.$$

We say that any  $\bar{w} \in S^*$  interpolates the training data since  $w^T x_i = y_i \forall i$ . Notice that we don't assume  $y$  is noiseless.

Further, if  $\bar{w} \in S^*$ , then

$$(+) \quad S^* = \bar{w} + \text{Ker}(X). \quad (\text{why?})$$

Today, we study two simple questions

- (1) How fast is GD at minimizing  $(\heartsuit)$ ?
  - (2) Which solution does it pick among  $S^*$ ?
- ↑ Does it even converge to one? 

Interlude: PL after compositions with linear maps

We saw last class that PL + smoothness yield fast convergence of GD. Thus, we should seek for objectives satisfying PL whenever possible. But how do we identify such objectives? The next result gives us a simple rule.

Proposition (A): Suppose that  $f$  is differentiable,  $A \in \mathbb{R}^{m \times d}$ , and  $b \in \mathbb{R}^m$ .

Define

$$\varphi(x) := f(Ax + b).$$

$$\|A\|_{op} := \sup_{\|v\|=1} \|Av\|_2 \quad \downarrow \quad \underbrace{v^T A^T A v}_{v^T A A^T v}$$

The following hold.

1) If  $f$  is  $L$ -smooth, then  $\varphi$  is  $(L\|A\|_{op}^2)$ -smooth.

2) If  $f$  satisfies  $\mu$ -PL and  $A$  is onto, then  $\varphi$  satisfies  $(\mu \lambda_{\min}(AA^T))$ -PL.

Proof: Let's start with 1), by the chain rule,

$$(\otimes) \quad \nabla \varphi(w) = A^T \nabla f(Aw + b).$$

So, for  $w, v \in \mathbb{R}^d$ ,

$$\begin{aligned} \|\nabla \varphi(w) - \nabla \varphi(v)\| &\leq \|A^T (\nabla f(Aw + b) - \nabla f(Av + b))\| \\ &\leq \|A\|_{op} \|\nabla f(Aw + b) - \nabla f(Av + b)\| \\ &\leq L \|A\|_{op} \|A(w - v)\| \\ &\leq L \|A\|_{op}^2 \|w - v\|. \end{aligned}$$

Since  $A$  is onto, then  $A\mathbb{R}^d = \mathbb{R}^m$ , and so

$$\varphi^* = \inf_w f(Aw + b) = \inf_z f(z) = f^*.$$

Thus, by  $(\otimes)$  we have

$$\begin{aligned} \frac{1}{2} \|\nabla g(w)\|^2 &= \frac{1}{2} \nabla f(Aw + b)^T A A^T \nabla f(Aw + b) \\ &\geq \frac{\lambda_{\min}(AA^T)}{2} \|\nabla f(Aw + b)\|^2 \end{aligned}$$

$$\geq \mu \lambda_{\min}(AA^T) (\underbrace{f(Aw+b) - f^*}_{\varphi(w) - \varphi^*}). \quad \square$$

Exercise: Show that if  $A$  is not surjective, then  $\varphi$  might not satisfy PL.  $\neg$

## Back to least squares

We can leverage Theorem (44) from Lecture 2 to derive the next result, which solves our first question.

Corollary: Suppose we ran GD with step size  $\eta_k = 1/\|A\|_{\text{op}}^2$  for (B). Then, its iterates satisfy

$$\text{dist}(w_k, S^*) \leq \left(1 - \frac{\lambda_{\min}(AA^T)}{\lambda_{\max}(AA^T)}\right)^k \text{dist}(w_0, S^*).$$

Proof: Take  $f(z) = \frac{1}{2n} \|z\|^2$ ,  $b = -y$ ,

and  $A = X$ . It is easy to see that

$\varphi$  is  $(\frac{1}{n})$ -smooth and it satisfies  $(\frac{1}{n})$ -PL. Then applying Proposition

(A) in tandem with Theorem (44) from lecture 2, gives the result.  $\square$

## Implicit regularization

Whenever a problem is ill-posed (more than one solution) it is common to add a regularizer to promote structure. In this case it is natural to use ridge regression

↑ Sec HW 1

$$w_\alpha = \arg \min_w \frac{1}{2n} \|Xw - y\|^2 + \frac{\alpha}{2} \|w\|^2.$$

Uniqueness of  $w_\alpha$  follows by strong convexity. By first-order optimality conditions

$$\frac{1}{n} X^T(Xw_\alpha - y) + \alpha w_\alpha = 0$$

$\Leftrightarrow$

$$w_\alpha = (X^T X + n\alpha I)^{-1} X^T y.$$

This gives a set of solutions  $\{w_\alpha : \alpha > 0\}$  that approaches  $S^*$  as  $\alpha \downarrow 0$ .

Proposition (+): We have

$$\lim_{\alpha \downarrow 0} w_\alpha = w^+ := \min_{w \in S^+} \|w\|.$$

Proof: Let  $X = U \overset{\text{diag}(\sigma_1, \dots, \sigma_n)}{\Sigma} V^T$  be the SVD decomposition of  $X$ . Then,

$$\lim_{\alpha \downarrow 0} w_\alpha = \lim_{\alpha \downarrow 0} V \text{diag} \left( \frac{\sigma_i}{\sigma_i^2 + \alpha n} \right) U^T y$$

$$= V \Sigma^{-1} U^T y$$

$$= \underbrace{X^T (X X^T)^{-1}}_{X^+} y = w^+.$$

$X^+$  Moore-Penrose pseudoinverse.

Let's show that  $w^+ \perp \text{Ker } X$ . Let  $z \in \text{Ker } X$ ,

$$(w^+)^T z = y^T (X X^T)^{-1} X z = 0.$$

Let  $w \in S^+ \setminus \{w^+\}$ , then there exists a  $z \in \text{Ker } X$  such that  $w = \underbrace{w^+}_{(+)} + z$  and so

$$\|w\|^2 = \|w^+\|^2 + \|z\|^2 > \|w^+\|^2,$$

$\nearrow$  Pythagoras

which completes the proof. □

In turn, GD also favors a similar solution.

Theorem (Implicit bias): Consider (b) with  $\text{rank}(X) = n$  and  $d > n$ . Suppose we ran GD with stepsize  $\eta \in (0, 2\eta / \|X\|_{\text{op}}^2)$  and initial iterate  $w_0$ . Then, its iterates converge to

$$w_\infty = X^\top y + \Pi_{\text{Ker} X} w_0.$$

Further

↑ Orthogonal projection.

$$w_\infty = \arg \min_{w \in S^*} \|w - w_0\|_2^2.$$

Proof: Let's first show that the iterates converge. (Why cannot conclude it from what we had before?)

Let us define the residual

$$r_k = X w_k - y.$$

By construction

$$r_{k+1} = X w_{k+1} - y$$

$$= X \left( w_k - \frac{\eta}{n} X^\top (X w_k - y) \right) - y$$

$$\begin{aligned}
&= r_k - \eta \frac{XX^T}{n} r_k \\
&= \left( I - \eta \frac{XX^T}{n} \right) r_k \\
&= \left( I - \eta \frac{XX^T}{n} \right)^{k+1} r_0.
\end{aligned}$$

Notice that  $0 < \lambda_{\min}\left(\frac{XX^T}{n}\right) \leq \frac{1}{n} \|X\|_{\text{op}}^2$ ,  
then since  $\eta \in (0, 2n/\|X\|_{\text{op}}^2)$  we get

$$\rho := \|I - \eta \frac{XX^T}{n}\|_{\text{op}} = \max_j |1 - \eta \lambda_j(\frac{XX^T}{n})| < 1.$$

Therefore,

$$\|r_k\|_2 \leq \rho^k \|r_0\| \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Expanding  $w_k$  directly,

$$w_k = \underbrace{\Pi_{\text{range } X^T}}_{X^+X \text{ (why?)}} w_k + \Pi_{\text{ker } X} w_k$$

$$= X^+X w_k + \Pi_{\text{ker } X} w_k$$

To simplify the second term, notice that

$$w_k = w_{k-1} - \eta X^T r_{k-1}$$

$$= w_0 - \eta \sum_{j=1}^k X^T r_{j-1}$$

Thus,  $\Pi_{\ker X} w_k = \Pi_{\ker X} w_0$ . Hence

$$w_k = X^T X w_k + \Pi_{\ker X} w_0$$

$$= X^T (y + r_k) + \Pi_{\ker X} w_0$$

$$\rightarrow X^T y + \Pi_{\ker X} w_0 \quad \text{as } k \rightarrow \infty.$$

To show that  $w_\infty$  is a minimum norm element, we decompose any  $w$  and  $w_0$  as

$w = w^T + z$  and  $w_0 = \Pi_{\text{range } X^T} w_0 + \Pi_{\ker X} w_0$ .  
Thus, Pythagoras gives

$$\|w - w_0\|^2 = \|w^T - \Pi_{\text{range } X^T} w_0\|^2$$

$$+ \|z - \Pi_{\ker X} w_0\|^2$$

Minimized  
when  $z = \Pi_{\ker X} w_0$

$$\geq \|w^T - \Pi_{\text{range } X^T} w_0\|^2$$

$$= \|w_\infty - w_0\|^2,$$

completing the proof. □